Quantum Field Theory – Homework 1

Name: ______________________ **Due Date:** 2026.03.30

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Lorentz Covariance of the Dirac Equation (20 points)

Consider the Dirac equation for a free spinor field

$$(\gamma^\mu \partial_\mu - m)\psi(x) = 0.$$

Under a Lorentz transformation $x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$, the spinor field transforms as

$$\psi(x) \to \psi'(x) = S(\Lambda)\,\psi(\Lambda^{-1}x),$$

where the gamma matrices satisfy

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu.$$

- Show that if $\psi(x)$ satisfies the Dirac equation, then $\psi'(x)$ also satisfies

$$(\gamma^\mu \partial_\mu - m)\psi'(x) = 0.$$

  Conclude that the Dirac equation is **Lorentz covariant**.

Problem 2. Dirac Equation from the Dirac Action (20 points)

Consider the Dirac action for a free spinor field

$$S[\psi,\bar{\psi}] = \int d^4x\,\mathcal{L}, \qquad \mathcal{L} = -i\bar{\psi}(\gamma^\mu\partial_\mu - m)\psi.$$

- By varying the action with respect to $\bar{\psi}$, derive the Dirac equation

$$(\gamma^\mu\partial_\mu - m)\psi = 0.$$

By varying the action with respect to ψ, derive the adjoint Dirac equation

$$(\gamma^\mu\partial_\mu + m)\bar{\psi} = 0.$$

Problem 3. Lorentz Transformation of a Dirac Bilinear (20 points)

Consider the Dirac bilinear

$$j^\mu(x) = \bar{\psi}(x)\gamma^\mu\psi(x).$$

Under a Lorentz transformation $x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$, suppose the spinor and its adjoint transform as

$$\psi(x) \to \psi'(x) = S(\Lambda)\psi(\Lambda^{-1}x), \qquad \bar{\psi}(x) \to \bar{\psi}'(x) = \bar{\psi}(\Lambda^{-1}x)S^{-1}(\Lambda),$$

and assume that

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu\gamma^\nu.$$

- Show that the bilinear $j^\mu(x) = \bar{\psi}(x)\gamma^\mu\psi(x)$ which is interpreted as the Dirac current transforms as a Lorentz four-vector, namely

$$j'^\mu(x) = \Lambda^\mu{}_\nu\, j^\nu(\Lambda^{-1}x).$$

Problem 4. Infinitesimal Lorentz Transformations and Gamma Matrices (20 points)

The gamma matrices satisfy the transformation law

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu.$$

For an infinitesimal Lorentz transformation, write

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \qquad S(\Lambda) = 1 + \frac{1}{2}\omega_{\rho\nu}\Sigma^{\rho\nu},$$

where $\omega_{\rho\nu} = -\omega_{\nu\rho}$ is infinitesimal, and $\Sigma^{\rho\nu}$ are the generators of Lorentz transformations in the spinor representation.

- Expand the relation

  $$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu$$

  to first order in $\omega_{\rho\nu}$, and show that the generators satisfy

  $$i[\Sigma^{\lambda\rho}, \gamma^\mu] = \eta^{\lambda\mu}\gamma^\rho - \eta^{\rho\mu}\gamma^\lambda.$$

- The matrices $\Sigma^{\rho\nu}$ generating Lorentz transformations on spinors are therefore required to satisfy

  $$i[\Sigma^{\rho\nu}, \gamma^\mu] = \eta^{\rho\mu}\gamma^\nu - \eta^{\nu\mu}\gamma^\rho.$$

  Verify that the choice

  $$\Sigma^{\rho\nu} = \frac{i}{4}[\gamma^\rho, \gamma^\nu]$$

  indeed satisfies this commutation relation. Conclude that this gives the generators of the spinor representation of the Lorentz group.